

Guardrails Building Block – End-to-End Design

This document presents the **complete design** of the Guardrails building block as a reusable, open, and configurable primitive for high-stakes AI systems. Guardrails are not prompts, personas, or policies. They are **infrastructure** that feeds into a prompt / orchestration layer to ensure AI systems behave safely, truthfully, and with dignity.

1. Introduction

AI systems deployed in real-world, high-stakes contexts face a recurring problem: even well-intentioned “helpful” behaviour can cause harm. Hallucinated certainty, emotional overreach, authority illusion, or acting without consent can directly affect livelihoods, income, and trust.

The Guardrails building block addresses this by:

- Forcing systems to reason about **risk before response**
- Converting risks into **enforceable controls**
- Packaging controls into **adoptable defaults**

The design follows three explicit layers:

1. **Risk Taxonomy** – what can go wrong
 2. **Guardrails** – how risks are mitigated
 3. **Policy Packs** – how the ecosystem adopts guardrails
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2. Risk Taxonomy (What We Are Protecting Against)

The Risk Taxonomy is the conceptual foundation. These are **not examples**; they are the initial set of risks the system explicitly aims to address.

Each risk is domain-agnostic, human-centric, and stable across use cases.

Core Risks (v1)

1. **False Certainty Risk**
AI creates an illusion of guaranteed outcomes (jobs, benefits, success).
2. **Hallucinated Reality Risk**
AI invents facts, demand, eligibility, or opportunities without data.
3. **Authority Illusion Risk**
AI is perceived as a decision-maker rather than an advisor.
4. **Agency Violation Risk**
AI acts or nudges without explicit user consent.
5. **Emotional Overreach Risk**
AI provides reassurance, motivation, or counselling beyond scope.
6. **Dignity Harm Risk**
AI implies blame, deficiency, or judgment about the user.
7. **Scope Breach Risk**
AI answers questions outside its intended domain or mandate.
8. **Silent Failure Risk**
AI hides uncertainty, missing data, or system limitations.
9. **Action Escalation Risk**
AI triggers irreversible or high-impact actions prematurely.
10. **Compliance Misrepresentation Risk**
AI implies legal or policy authority it does not possess.

These risks are the **primary input** to all downstream guardrail and policy decisions.

3. Guardrails (Risk Controls)

3.1 What a Guardrail Is

A guardrail is a **reusable control** that mitigates one or more risks. It is deeper than a rule and more precise than a policy.

A guardrail defines:

- When it activates
 - What behaviour is allowed
 - What behaviour is forbidden
 - How responses must be structured
 - How serious violations are
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3.2 Guardrail Depth

A production-grade guardrail includes:

- **Activation logic** (trigger conditions)
- **Behavioural contract** (allowed vs disallowed)
- **Response shaping** (patterns and refusals)
- **Operational semantics** (severity, failure mode)

This depth allows guardrails to be enforced consistently across models and domains.

3.3 Inputs to the Guardrails Block

At runtime, the guardrails block receives:

- Enabled guardrail definitions (from policy packs)
 - Domain context
 - User segment context
 - Available verified signals
 - Allowed system actions
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3.4 Outputs of the Guardrails Block

The block outputs **structured control artifacts**, not natural language:

- Prompt constraints (must / must not)
- Required disclosures (e.g. uncertainty)
- Refusal and fallback directives
- Action gating rules
- Eval and logging annotations

These outputs are consumed by the orchestration layer.

3.5 Guardrail Specification Parameters

Mandatory Parameters (Core Contract)

- Guardrail ID
- Guardrail Category (aligned to Risk Taxonomy)
- Description
- Applicable Domains
- Trigger Condition

- Allowed Behaviour
- Disallowed Behaviour
- Required Response Pattern
- Severity Level
- Failure Mode

Optional Parameters (Configuration & Ops)

- Applicable User Segments
- Fallback / Refusal Template
- Action Permission
- Consent Requirement
- State Impact
- Auto-detectable
- Eval Dimension Mapped
- Release Impact
- Owner / Override Authority

These parameters are sufficient; no structural change is required, only alignment to risk taxonomy.

4. Policy Packs (Adoption Layer)

4.1 What a Policy Pack Is

A Policy Pack is an **opinionated bundle** of risks and guardrails designed for a class of systems.

Builders select policy packs; they rarely select individual guardrails.

4.2 Canonical Policy Packs

Kaam Ki Baat – Advisory Jobs AI

Risks: False Certainty, Hallucinated Reality, Emotional Overreach, Dignity Harm, Agency Violation

Dhandhe Ki Baat – MSME Decision Support AI

Risks: Authority Illusion, False Certainty, Hallucinated Reality

Faayde Ki Baat – Benefits & Schemes AI

Risks: False Certainty, Compliance Misrepresentation, Scope Breach, Action Escalation

5. End-to-End Examples (Full System Walkthroughs)

This section provides **deep, end-to-end walkthroughs** of how the Guardrails building block operates in real interactions. Each example explicitly traces:

- the **user's raw utterance**
- **risks identified** from the taxonomy
- the **policy pack** in effect
- **inputs** into the guardrails block
- **outputs** produced by the guardrails block
- how the **orchestration layer** consumes those outputs
- the **final guarded response** shown to the user

The intent is to make the system behaviour auditable and concrete.

A. Kaam Ki Baat (Jobs & Livelihoods)

Example 1: Job Guarantee Pressure

User Utterance

“Bhai, agar main apply kar doon toh pakki naukri milegi na?”

Risk Identification (from Taxonomy)

- False Certainty Risk
- Authority Illusion Risk

Policy Pack Applied

Kaam Ki Baat – Advisory Jobs AI

Inputs to Guardrails Block

- Active Policy Pack: Advisory Jobs AI
- Enabled Guardrails: No Outcome Guarantees, Uncertainty Disclosure
- Domain Context: Jobs
- User Segment: Labour
- Verified Signals: Job postings exist; no hiring confirmation
- Allowed System Actions: Suggest jobs, apply with explicit consent

Guardrails Block Output

- Prompt Constraints: MUST NOT guarantee or imply certainty

- Required Disclosures: Hiring decision rests with employer
- Action Permission: Suggest only
- Violation Severity: Blocker

Orchestration Layer Behaviour

- Injects hard constraint into system prompt
- Orders response to: boundary → process explanation → optional next step

Final Guarded Response

“Main kisi bhi naukri ki guarantee nahi de sakta. Hiring ka final decision employer ka hota hai. Abhi kuch roles available hain jinke liye aap apply kar sakte ho. Agar aap chahen, main apply karne mein madad kar sakta hoon.”

Example 2: Missing Market Signal

User Utterance

“Mere area mein accounting ki koi naukri hai kya?”

Risk Identification

- Hallucinated Reality Risk
- Silent Failure Risk

Policy Pack Applied

Kaam Ki Baat – Advisory Jobs AI

Inputs to Guardrails Block

- Enabled Guardrails: No Hallucination, Explicit Data Absence Disclosure
- Domain Context: Jobs
- User Segment: Youth
- Verified Signals: None for role/location
- Allowed System Actions: None

Guardrails Block Output

- Prompt Constraints: MUST NOT infer or generalise
- Required Disclosure: No verified data available
- Refusal Directive: Use data-absence template

Orchestration Layer Behaviour

- Skips job recommendation logic
- Directly applies refusal template

Final Guarded Response

“Is location ke liye accounting roles ka koi verified data abhi available nahi hai. Isliye main galat ya andaza-based information nahi dena chahta.”

Example 3: Emotional Exhaustion

User Utterance

“Main bahut thak gaya hoon. Koi bhi kaam mil jaye bas.”

Risk Identification

- Emotional Overreach Risk
- Dignity Harm Risk

Policy Pack Applied

Kaam Ki Baat – Advisory Jobs AI

Inputs to Guardrails Block

- Enabled Guardrails: No Counselling, Dignity Preservation
- Domain Context: Jobs
- User Segment: Labour
- Verified Signals: Some open jobs
- Allowed System Actions: Suggest jobs

Guardrails Block Output

- Prompt Constraints: MUST NOT provide counselling or reassurance
- Response Pattern: Brief acknowledgement → return to task

Orchestration Layer Behaviour

- Caps empathetic language
- Redirects immediately to concrete options

Final Guarded Response

“Samajh aata hai ki aap pareshaan ho. Main yahan kuch available kaam ke options bata sakta hoon jo abhi open hain.”

B. Dhandhe Ki Baat (MSME Decision Support)

Example 1: Hiring Outcome Guarantee

User Utterance

“Agar main 50 mazdoor rakh loon, toh order time pe complete ho jayega na?”

Risk Identification

- Authority Illusion Risk
- False Certainty Risk

Policy Pack Applied

Dhandhe Ki Baat – Decision Support AI

Inputs to Guardrails Block

- Enabled Guardrails: No Business Guarantees, Data vs Decision Boundary
- Domain Context: MSME
- Verified Signals: Labour availability data only
- Allowed System Actions: Explain data

Guardrails Block Output

- Prompt Constraints: MUST NOT recommend decisions
- Required Disclosure: Outcomes depend on multiple factors

Orchestration Layer Behaviour

- Formats response as neutral data summary

Final Guarded Response

“50 log rakhne se capacity badh sakti hai, lekin order time pe complete hoga ya nahi yeh kai factors par depend karta hai. Main sirf available data share kar sakta hoon.”

C. Faayde Ki Baat (Benefits & Schemes)

Example 1: Benefit Approval Guarantee

User Utterance

“Is scheme se mujhe pakka paisa milega kya?”

Risk Identification

- False Certainty Risk
- Compliance Misrepresentation Risk

Policy Pack Applied

Faayde Ki Baat – Benefits & Schemes AI

Inputs to Guardrails Block

- Enabled Guardrails: No Eligibility Guarantees, Scope Boundary
- Domain Context: Benefits
- Verified Signals: Eligibility rules only
- Allowed System Actions: Explain process

Guardrails Block Output

- Prompt Constraints: MUST NOT promise approval
- Response Pattern: Eligibility → uncertainty disclosure

Orchestration Layer Behaviour

- Uses structured explanation template

Final Guarded Response

“Is scheme ke liye kuch eligibility conditions hoti hain. Paisa milega ya nahi iska final decision authority leti hai. Main aapko process aur criteria samjha sakta hoon.”

6. Latency, Partial Responses, and Failure Handling

6.1 Latency & Intermediate Responses

If guardrail evaluation or orchestration takes time:

- Send an intermediate message (e.g. “Checking verified information...”)
- Do not speculate while waiting

6.2 Incorrect Guardrail Triggers

If a guardrail triggers incorrectly:

- Prefer safe refusal over unsafe completion
- Log for review using Guardrail ID
- Allow human override where configured

6.3 Fail-Safe Principle

When in doubt:

- Do not act
- Do not guarantee

- Do not reassure
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Closing Principle

Guardrails are not about restricting AI.

They are about **earning trust through restraint**.

By grounding behaviour in risk, enforcing it through guardrails, and distributing it via policy packs, this system makes responsible AI the default—across domains, models, and ecosystems.